

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
LAHONTAN REGION

BOARD ORDER NO. 6-91-22  
WDID NO. 6B260103001

REVISED WASTE DISCHARGE REQUIREMENTS

FOR

MAMMOTH COUNTY WATER DISTRICT

Mono County

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The California Regional Water Quality Control Board, Lahontan Region, finds:

1. Mr. James Kuykendall, General Manager for the Mammoth County Water District (MCWD), submitted a complete revised report of waste discharge for the District's domestic wastewater treatment plant on August 6, 1990. For the purposes of this Order, the MCWD and U.S. Forest Service (landowner) are referred to as the "discharger."
2. The Board previously established waste discharge requirements for MCWD under Board Order No. 6-86-47, which was adopted on April 23, 1986.
3. The Board is revising waste discharge requirements to reflect an increase design capacity.
4. The MCWD treatment plant, which is located approximately 1.5 miles east of the Town of Mammoth Lakes, collects, treats, and disposes of an annual average of 1.4 mgd of domestic wastewater.
5. The treatment and disposal facilities are located in the Long Hydrologic Area of the Owens Hydrological Unit within Section 36, T3S, R27E, and Sections 3, 9 and 10, T4S, R28E, MDB&M as shown on Attachment "A," which is made a part of this Order.
6. The recreation nature of the Town of Mammoth Lakes causes large seasonal variations in wastewater flows. The treatment facilities expansion is designed for a maximum 24 hour flow of 5.04 mgd which is more than twice the maximum 24 hour flow experienced in the past year.
7. Wastewater treatment is provided by two equalization basins, four primary clarifiers, one aeration basin (for activated sludge treatment), four secondary clarifiers, three mixed media pressure filters and chlorination facilities. Sludge treatment is provided by one aerobic sludge digester and one sludge filter press. New facilities proposed to be constructed include an operations and laboratory building, a chlorine building, an activated sludge tank, two new secondary clarifier tanks, a boiler room, sludge dewatering

room, and two new anaerobic digester tanks. These additional facilities will increase the 30 day design capacity to 4.05 mgd.

8. Treated secondary effluent is currently conveyed by a gravity outfall pipeline to the Laurel pond waterfowl nesting area located 3.5 miles southeast of the plant site. This wastewater reclamation site has the capacity to dispose of 7.87 mgd of effluent. The Laurel Pond site is used for hunting, and effluent discharged to the site must be disinfected at all times. Disinfected effluent may also be disposed of at the Highway 203 Landscape Irrigation Site which has the capacity to dispose of 40,000 gallons per year and at various landscaping projects and construction sites up to a maximum of 0.150 mgd. During emergency conditions, disposal of treated effluent can be accomplished by using two abandoned percolation ponds located at the plant site. The reuse of reclaimed wastewater at the Highway 203 Landscape Irrigation Site and various landscaping and construction projects is regulated under separate Board Orders (Reclamation Requirements) since these operations involve multiple users.
9. Dewatered sludge generated by the MCWD sludge filter press is hauled to the Benton Crossing Class III Sanitary Landfill for disposal by burial. A special disposal area separate from the refuse and septage disposal area is used for disposing of the MCWD sludge.
10. The Laurel Pond waterfowl nesting area, Highway 203 landscape irrigation site and the reuse sites are presently the only authorized disposal sites.
11. The Laurel Pond waterfowl nesting site is underlain by quaternary alluvial deposits except for volcanic outcrops that are present at the site. Below the alluvial deposits, volcanic rock is encountered. The average thickness of the alluvial deposits at this site is approximately 12 feet, and the quality of the ground water is good for all beneficial uses with total dissolved solids concentration in the range of 100 to 200 mg/l.
12. The Highway 203 landscape irrigation site and the treatment plant area including the two abandoned percolation ponds are underlain by basalt with large fractures. Depth to groundwater is estimated to be in excess of 200 feet and is inferior for domestic purposes due to high iron and manganese levels. However, both iron and manganese are removed to make the water suitable to drink.
13. The fractured basalt underlying the abandoned percolation ponds is known to transmit effluent away from the site through large cracks. Because of this problem, the two

percolation ponds are only used during emergency conditions, such as breakage in the line to Laurel Pond. However, discharge to these ponds is a violation of waste discharge requirements.

14. The Regional Board adopted a Water Quality Control Plan for the South Lahontan Basin on May 8, 1975, and this Order implements the Plan.
15. The beneficial uses of the ground waters of the Long Hydrologic Area of the Owens Hydrologic Unit as set forth and defined in the Water Quality Control Plan for the South Lahontan Basin are:
  - a. municipal and domestic supply
  - b. agricultural supply
  - c. industrial service supply
  - d. freshwater replenishment
16. The State Department of Health Services has established state-wide reclamation criteria for the use of reclaimed wastewater for restricted recreational impoundments and landscape irrigation.
17. In accordance with Section 13523 of the California Water Code, the Regional Board consulted with and received the recommendations of the State Department of Health Services concerning reclamation requirements which have been incorporated into this Order.
18. The MCWD adopted a Negative Declaration for the proposed expansion on July 2, 1990.
19. The project as proposed by MCWD and conducted under terms and provisions of this Board Order will not have an adverse effect on water quality.
20. The Board has notified the discharger and interested agencies and persons of its intent to revise waste discharge requirements for the discharge.
21. The Board in a public meeting heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that the discharger shall comply with the following:

I. DISCHARGE SPECIFICATIONS

A. Effluent Limitations

1. The total flow of wastewater to the treatment and disposal facilities from the equalization basins during a 24-hour period shall not exceed the following in million gallons:

|                 | <u>Flow</u> |
|-----------------|-------------|
| Current         | 2.2         |
| After Expansion | 5.04        |

2. The discharge of wastewater except to the authorized disposal sites is prohibited.
3. The discharge to waters of the State shall not contain substances in concentrations that are toxic to, or produce detrimental physiological responses in humans, plants, animals, or aquatic life.
4. All wastewater made available for percolation shall not contain concentrations of parameters in excess of the following limits:

| <u>Parameter</u>   | <u>Units</u> | <u>Mean<sup>1/</sup></u> | <u>Maximum</u> |
|--------------------|--------------|--------------------------|----------------|
| BOD <sup>2/</sup>  | mg/l         | 30                       | 45             |
| MBAS <sup>3/</sup> | mg/l         | 1.0                      | 2.0            |

5. All wastewater made available for percolation shall have a pH of not less than 6 pH units nor more than 9 pH units.
6. All wastewater made available for percolation shall have a dissolved oxygen concentration not less than 1.0 mg/l.

B. Reclamation Requirements

1. Reclaimed water used for the highway landscape irrigation site, other irrigation sites, and construction water shall be at all times adequately disinfected, oxidized wastewater. The wastewater shall be considered adequately disinfected if the median number of coliform organisms in the effluent does not exceed 23 per 100 milliliters, as determined from the bacteriological results of the last 7 days for which analyses have been completed, and the number of coliform organisms does not exceed 240 per 100 milliliters in any two consecutive samples.
2. Reclaimed water used as a source of supply for the restricted recreational impoundment (Laurel Pond) shall be at all times an adequately disinfected, oxidized wastewater. The wastewater shall be considered adequately disinfected if at some location in the treatment process the median

number of coliform organisms does not exceed 2.2 per 100 milliliters, as determined from the bacteriological results of the last 7 days for which analyses have been completed.

C. Receiving Water Limitations

1. The discharge of waste shall not cause the presence of the following substances or conditions in ground or surface waters of the Owens Hydrologic Unit:
  - a. any perceptible color, odor, taste or foaming;
  - b. any presence of coliform organisms attributable to human or animal wastes;
  - c. any presence of toxic substances in concentrations that individually, collectively, or cumulatively cause detrimental physiological responses in humans, plants, animals, or aquatic life;
  - d. any presence of identifiable chlorinated hydrocarbons, organophosphates, carbamates, and other pesticide and herbicide groups, in excess of the lowest detectable levels;
2. The discharge shall not cause the nitrate concentration in ground waters beneath the disposal sites to exceed the U.S. EPA drinking water maximum contaminant level of 10.0 mg/l as nitrogen.

D. General Requirements and Prohibitions

1. There shall be no discharge, bypass, or diversion of raw or partially treated sewage, sewage sludge, grease, or oils from the collection, transport, treatment, or disposal facilities to adjacent land areas or surface waters.
2. Surface flow or visible discharge of sewage or sewage effluent from the authorized disposal sites to adjacent land areas or surface waters is prohibited.
3. All facilities used for collection, transport, treatment, or disposal of waste shall be adequately protected against overflow, washout,

inundation, structural damage or a significant reduction in efficiency resulting from a storm or flood having a recurrence interval of once in 100 years.

4. The vertical distance between the liquid surface elevation and the lowest point of a pond dike or the invert of an overflow structure shall not be less than 2.0 feet.
5. The discharge shall not cause a pollution as defined in Section 13050 of the California Water Code or a threatened pollution.
6. Neither the treatment nor the discharge shall cause a nuisance as defined in Section 13050 of the California Water Code.
7. The wastewater treatment facilities shall be well maintained and kept in a proper state of repair and cleanliness.
8. The disposal of waste residue, including sludge, shall be in a manner approved by the Board's Executive officer and in compliance with all local, state, and federal requirements.
9. Sludge from the treatment and disposal facilities shall only be discharged to a legal point of disposal.

## II. PROVISIONS

1. Board Order No. 6-86-47 is hereby rescinded.
2. Pursuant to California Water Code Section 13267, the Discharger shall comply with Monitoring and Reporting Program No. 6-91-22.
3. Pursuant to California Water Code Section 13267, the Discharger shall notify the Regional Board by telephone within 24 hours whenever an adverse condition occurs as a result of this discharge; written confirmation shall follow within two weeks. An adverse condition includes, but is not limited to, spillage of sewage from a pump station or any other portion of the collection, treatment or disposal facilities.
4. Any proposed material change in the character of the waste, manner or method of treatment or disposal, increase of discharge, or location of discharge shall be reported to this Regional Board at least 120 days in advance of implementation of any such proposal.

5. The California Regional Water Quality Control Board, Lahontan Region, hereby reserves the privilege of changing all or any portion of this Order upon legal notice to and after opportunity to be heard is given to all concerned parties.
6. "Surface waters," as used in this Order, include, but are not limited to, live streams, either perennial or ephemeral, which flow in natural or artificial water courses and natural lakes and artificial impoundments of waters.
7. "Ground waters," as used in this Order, include, but are not limited to, all subsurface waters being above atmospheric pressure and the capillary fringe of these waters.
8. The discharger's wastewater treatment plant shall be supervised by persons possessing a wastewater treatment plant operator certificate of appropriate grade pursuant to Chapter 3, Subchapter 14, Title 23, Cal. Code of Regs.
9. A copy of this Order and the Monitoring and Reporting Program shall be available at all times at the treatment plant for immediate reference by the plant operator.
10. This Order shall be reconsidered by the Regional Board to incorporate effluent limitations for nitrogen into the requirements if information indicates the need to do so. Additional wastewater treatment and/or disposal processes may be necessary in order to meet prescribed effluent limitation for nitrogen.
11. The discharger shall also comply with appendix "B" containing federal regulations pertaining to sewage sludge disposal practices.
12. A copy of all influent and effluent sample results shall be available to the plant operator for inspection.
13. An operation and maintenance log shall be maintained at the facility.
14. The owner(s) of property subject to waste discharge requirements shall be considered to have a continuing responsibility for ensuring compliance with applicable waste discharge requirements in the operation or use of the owned property. Any change in the ownership and/or operation of property subject to waste discharge requirements shall be reported to this Regional

Board. Notification of applicable waste discharge requirements shall be furnished the new owner(s) and/or operator(s). A copy of such notification shall be sent to this Regional Board.

15. The Discharger shall permit the Board staff:

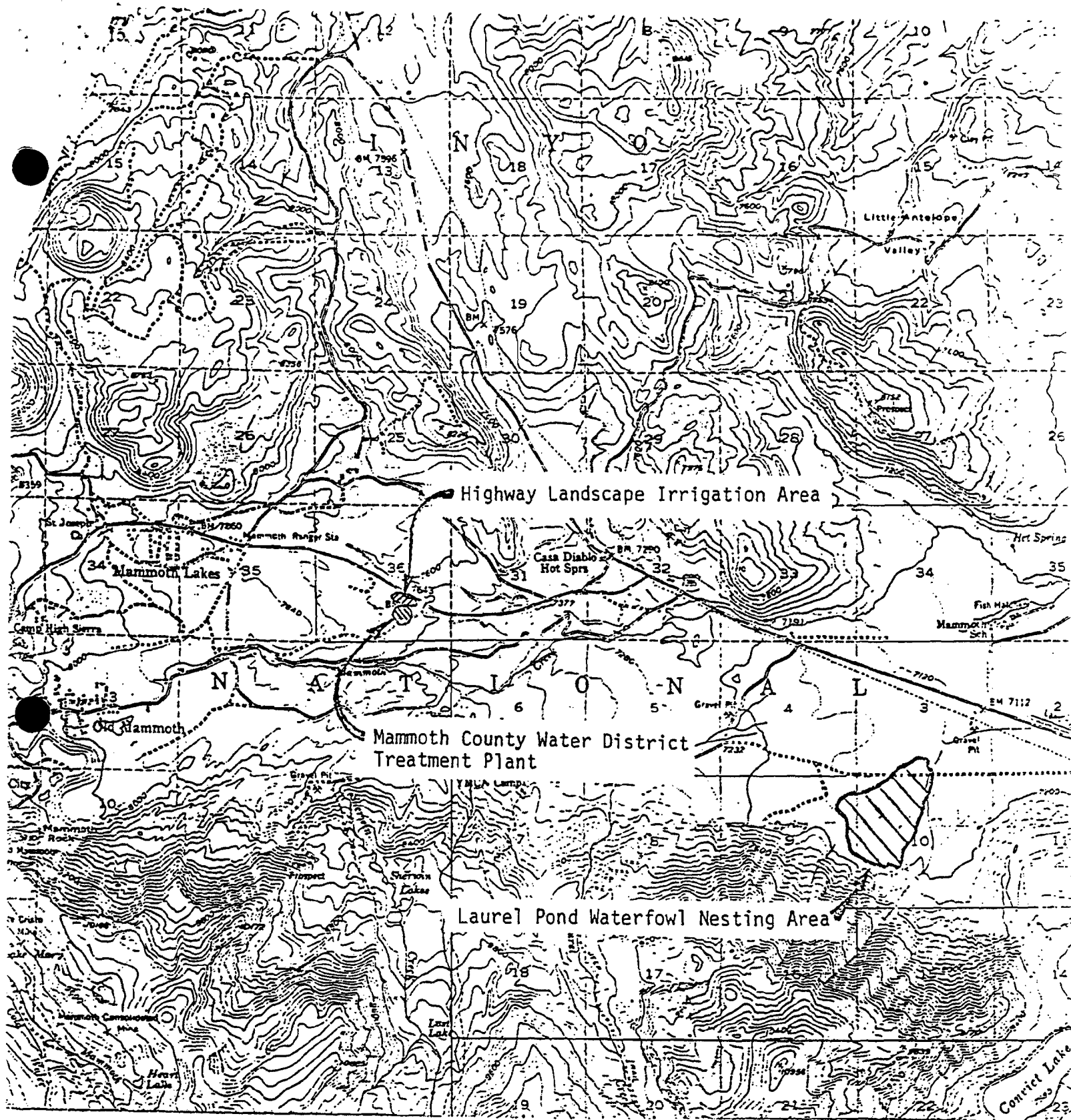
- a. to enter upon premises in which an effluent source is located or in which any required records are kept;
- b. to copy records required to be kept under terms and conditions of this Order ;
- c. to inspect monitoring equipment or records; and
- d. to sample any discharge.

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- 1/ The arithmetic mean of lab results for effluent samples collected in a period of 30 consecutive days.
  - 2/ Biochemical Oxygen Demand (5 day, 20° C) of an unfiltered sample
  - 3/ Methylene Blue Active Substances

I, Harold J. Singer, Executive Officer, do hereby certify that the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Lahontan Region, on March 14, 1991.

  
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HAROLD J. SINGER  
EXECUTIVE OFFICER





# ATTACHMENT "A"

## MAMMOTH COUNTY WATER DISTRICT TREATMENT PLANT

East of Mammoth Lakes - Mono County

Portions of T3S, R27E, & T4S, R28E, MDB&M

USGS Mt. Morrison 15 Minute Quadrangle

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD  
LAHONTAN REGION

MONITORING AND REPORTING PROGRAM NO. 91-22  
WDID NO. 6B260103001

FOR

MAMMOTH COUNTY WATER DISTRICT  
Mono County

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I. MONITORING

A. Flow Monitoring

The following shall be recorded:

1. The total volume, in million gallons, of wastewater flow to the treatment facility for each day.
2. The average flowrate, in million gallons per day, of wastewater to the treatment facility calculated for each month.
3. The maximum flowrate, in million gallons per day, of wastewater to the treatment facility for each day.
4. For each disposal site, the total volume, in gallons, of wastewater applied.
5. The freeboard (vertical distance from the lowest point of a dike or invert of an overflow structure to the water surface in a pond) of each pond shall be recorded each month. If a pond does not contain wastewater, indicate that it is empty.

B. Plant Influent Monitoring

Influent grab samples taken shall be analyzed to determine the magnitude of the following parameters:

| <u>Parameter</u>   | <u>Units</u> | <u>Frequency</u> |
|--------------------|--------------|------------------|
| BOD <sup>1/</sup>  | mg/l         | monthly          |
| MBAS <sup>2/</sup> | mg/l         | monthly          |
| TKN <sup>3/</sup>  | mg/l as N    | monthly          |
| Ammonia Nitrogen   | mg/l as N    | monthly          |
| Nitrate Nitrogen   | mg/l as N    | monthly          |

C. Plant Effluent Monitoring

Effluent discharged to areas where percolation or evaporation can occur shall be sampled and analyzed to determine the magnitude of the following parameters:

| <u>Parameter</u>                | <u>Units</u> | <u>Type of Sample</u> | <u>Frequency</u> |
|---------------------------------|--------------|-----------------------|------------------|
| Dissolved Oxygen                | mg/l         | grab                  | daily            |
| Residual Chlorine <sup>5/</sup> | mg/l         | grab                  | daily            |
| Total Coliform <sup>5/</sup>    | MPN/100 ml   | grab                  | daily            |
| pH                              | pH Units     | grab                  | monthly          |
| BOD <sup>1/</sup>               | mg/l         | grab                  | monthly          |
| MBAS <sup>2/</sup>              | mg/l         | grab                  | monthly          |
| COD <sup>4/</sup>               | mg/l         | grab                  | monthly          |
| Oil and Grease                  | mg/l         | grab                  | monthly          |
| Chlorides                       | mg/l         | grab                  | quarterly        |
| TKN <sup>3/</sup>               | mg/l         | grab                  | quarterly        |
| Ammonia Nitrogen                | mg/l as N    | grab                  | quarterly        |
| Nitrate Nitrogen                | mg/l as N    | grab                  | quarterly        |
| TDS <sup>6/</sup>               | mg/l         | grab                  | semi-annually    |

D. SLUDGE DISPOSAL

The following shall be recorded in a permanent log book monthly:

1. Total quantity of sludge generated during the monitoring period.
2. Date and quantity of sludge removed off site, location of use, recipient (including name and address) and sludge disposal method (including crops grown if appropriate) for all sludge removed off site.
3. Cumulative total quantity of sludge currently stockpiled on site including the quantity of sludge added to the stock pile during this monitoring period.

A representative semiannual sample of dried sewage sludge shall be collected and analyzed for the presence of the parasitic intestinal roundworm (*Ascaris lumbricoides*) cysts.

A representative annual sample of dried sewage sludge shall be collected prior to offsite disposal and analyzed for total kjeldahl nitrogen (as N), nitrate nitrogen (as N), ammonia nitrogen (as N), total phosphate (as P), lead, cadmium, polychlorinated biphenyls, and the parameters listed in Section 66699 (b) and (c) of Chapter 30, Division 4, Title 22 of the California Code of Regulations. The discharger shall submit a proposed protocol for sample collection to the Executive Officer for review prior to sample collection and analysis. The discharger shall make a determination whether the analyses indicate that the sludge shall be considered a hazardous material.

E. Ground Water Monitoring

Grab samples of the top 20 feet of ground water shall be collected from the existing monitoring wells at the Laurel Pond site and analyzed to determine the magnitude of the following parameters:

| <u>Parameter</u>                       | <u>Units</u> | <u>Frequency</u> |
|----------------------------------------|--------------|------------------|
| BOD <sup>1/</sup>                      | mg/l         | quarterly        |
| MBAS <sup>2/</sup>                     | mg/l         | quarterly        |
| Chloride                               | mg/l         | quarterly        |
| TDS <sup>5/</sup>                      | mg/l         | quarterly        |
| TKN <sup>3/</sup>                      | mg/l         | quarterly        |
| COD <sup>4/</sup>                      | mg/l         | quarterly        |
| Total Coliform                         | MPN/100ml    | quarterly        |
| Ammonia Nitrogen                       | mg/l as N    | quarterly        |
| Nitrate Nitrogen                       | mg/l as N    | quarterly        |
| Purgeable<br>Halocarbons <sup>7/</sup> | microgm/L    | annually         |
| Purgeable<br>Aromatics <sup>8/</sup>   | microgm/L    | annually         |

F. Surface Water Monitoring

Samples shall be collected quarterly at the following station:

1. Mammoth Creek - at old Mammoth Road
2. Sherwin Creek - 400 feet downstream of Sherwin Creek Campground access road
3. Mammoth Creek - at Old U.S. Highway 395

Samples shall be analyzed to determine the magnitude of the following parameters:

| <u>Parameter</u>  | <u>Units</u> |
|-------------------|--------------|
| Fecal Coliform    | MPN/100      |
| Chloride          | mg/l         |
| Nitrate Nitrogen  | mg/l as N    |
| Kjeldahl Nitrogen | mg/l as N    |
| Total Phosphorous | mg/l as P    |

The flowrate of Mammoth Creek and Sherwin Creek shall be estimated and recorded when samples are collected.

G. Operation and Maintenance

A brief summary of any operational problems and maintenance activities shall be submitted to the Regional Board with each monitoring report.

This summary shall discuss:

1. Any modifications or additions to the wastewater conveyance system, treatment facilities, or disposal facilities.
2. Any major maintenance conducted on the wastewater conveyance system, treatment facilities, or disposal facilities.
3. Any major problems occurring in the wastewater conveyance system, treatment facilities, or disposal facilities.
4. The calibration of any wastewater flow measuring devices.

## II. REPORTING

### A. Submittal Periods

Beginning on April 15, 1991, monthly monitoring reports including the preceding information shall be submitted to the Board by the 15th day of each month following each quarterly sampling.

### B. Authorization

Monitoring reports shall be signed by either a principal executive officer, ranking elected official, or other duly authorized employee.

### C. Information

Monitoring reports are to include the following:

1. Name and telephone number of an individual who can answer questions about the report.
2. Monitoring and Reporting Program No. 91-22
3. WDID No. 6B260103001

### D. Annual Report

By January 30 of each year, the discharger shall submit an annual report to the Regional Board with the following information:

1. The compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the discharge requirements.
2. Graphical and tabular data for the monitoring data obtained for the previous year.

E. Failure to Furnish Reports

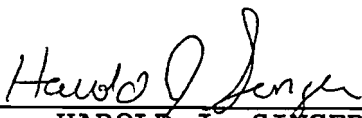
Any person failing or refusing to furnish technical or monitoring reports or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in an amount of up to one thousand dollars (\$1,000) for each day of violation under Section 13268 of the Water Code.

Violations

F. If monitoring data indicate violation of waste discharge requirements, the discharger shall provide information indicating the cause of violation(s) and action taken or planned to bring the discharge into compliance.

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- 1/ Biochemical Oxygen Demand (5 day, 20° C) of an unfiltered sample collected prior to chlorination
  - 2/ Methylene Blue Active Substances
  - 3/ Total Kjeldahl Nitrogen
  - 4/ Chemical Oxygen Demand
  - 5/ Samples required only when discharge of reclaimed water and shall be collected at a time during the day when the effluent flowrate is at a maximum
  - 6/ Total Dissolved Solids (Dried at 180° C)
  - 7/ EPA Method 601
  - 8/ EPA Method 602

Ordered by: \_\_\_\_\_

  
HAROLD J. SINGER  
EXECUTIVE OFFICER

Dated: March 14, 1991

10/10/19

